Topic 4: Interaction Analysis with Apple Watch Data

December 2025

1 Introduction

Sleeping is crucial to be healthy. The quality of sleep is highly related to cardiovascular, metabolic, and mental well-being. For many years, sleep monitoring was done via polysomnography (PSG), so it required the people to sleep in a laboratory, making it expensive and not suitable for long periods.

Nowadays, the usage of wearable devices has been increasing, especially the smartwatches (Apple Watch). These devices can track many a wide range of health, fitness, and activity metrics, collecting data continuously. Thanks to this, we could provide more extended analysis of sleep or even approximate the sleep stages.

Some previous researchers also explored this field. For example, Dzierzewski [1] have demonstrated in their paper that lifestyle and activity data that reflects a good physical health are associated with deeper sleep. Other studies, like Boudreau [2] revealed that heart rate variability varies significantly during sleep stages, which could contribute directly to distinguishing deep sleep. A related study done by Dieli-Conwright [3], also demonstrates that the patterns in heart rate and activity are associated with deeper sleep. Furthermore, Walch [4] demonstrated that given the motion, heart rate and other variables, is possible to predict if the user is asleep or awake with up to 90% accuracy.

However, most studies aren't focused on predicting sleep stages, instead, they are analyzing and measuring sleep quality given some metrics and surveys, or they are only classifying sleep/wake. This gives an opportunity to go a step further and create a model that could distinguish between sleep stages.

Therefore, this project has two main goals. First, we will build a model that uses only Apple Watch data [5] to predict a person's sleep stages and given these classifications we will understand, from the model, which are the conditions that support deep sleep.

To do this, we will use a dataset collected from 31 people who wore an Apple Watch for 7-14 days and also had one night of sleep monitored in a laboratory with PSG. We cleaned the data, created helpful features from it, and used it to train many machine learning models, mainly explainable models like Gradient Boosting and Random Forest, in order to understand which are the key features for their predictions. Our results show that our model can successfully identify sleep stages with high accuracy.

2 Problem Formulation

The main challenge is that sleep monitoring often relies on basic sleep/wake classification, some other researches don't focus on classifying, instead they analyze and measure sleep quality. This prevents doctors from obtaining detailed insights into individual sleep stages.

Deep sleep is essential for recovery, yet it is still unclear which wearable-derived signals can accurately predict it. So the problems in research are:

1. Can Apple Watch data accurately predict all distinct sleep stages?
2. Which features are the key for identifying deep sleep (Stage N3)?

3 Dataset Description

3.1 Dataset Introduction

The dataset for this project was collected by Olivia Walch [5][4] and consists of multimodal data from 31 healthy participants (21 females and 10 males), that did not possess any REM sleep behavior disorder or obstructive sleep apnea.

The participants wore an Apple Watch in their normal daily life for 7-14 days. Then the last night each of the participants slept in a sleep laboratory where they also kept wearing the Apple Watch. So two things were recorded from each of the participants, the sleep stages from the polysomnography (PSG) test and all the metrics from the Apple Watch, which includes the heart rate (bpm), the acceleration (in g) in x, y and z, and the steps (count).

For each of the participants, each of the recorded data was stored in a different txt file. Those files are as follows:

- **motion (acceleration):** Each file is a user and contains the time (in seconds), and the accelerations (in g) in x, y and z axis. It has around 50 observations per second.[4]
- **heart rate (bpm):** Each file is a user and contains the time (in seconds) and the heart rates (in bpm (integer)) for each recorded moment. It has an observation every 5 to 10 seconds.[4]
- **steps (count):** Each file is a user and contains the time (in seconds) and the steps (integer) made for each recorded time. It has observations every 10 minutes.[4]
- **labeled sleep:** It contains the sleeping stage for each timestamp (recorded every 30 seconds). These stages are from 0 to 5, and they mean the following: wake = 0, N1 = 1, N2 = 2, N3 = 3, N4 = 4, REM = 5.[4]

3.2 Data Preprocessing

The goal of our preprocessing was to transform raw data from Apple Watch into a clean, well-adjusted, and usable dataset. We built the preprocessing upon the work of

Walch [4] (available in their github repository [6]), where we added an extra output feature. Then we proceeded to make some of our own preprocessing and feature engineering.
The preprocessing of Walch [4] consist of the following:

1. Read the raw data from the labeled sleeps and crop the timestamps into a range where the participant was about to fall asleep until a few seconds after the participant woke up. With this, we obtain the actual range of time that the participant slept.

2. Read the raw data from heart rate and acceleration and crop the data within that time range.

3. They create "Activity count" variable using the acceleration data. This was conducted by convolving the acceleration window with a Gaussian filter ($\sigma =$ 50 seconds) to smooth and aggregate the raw signal into counts. They used the code from te Lindert and Van Someren (2013) [7].

4. They align the number of timestamps of the cropped data (heart rate and Activity count) with the labeled sleeps by making an interpolation of the data.

5. They create a variable named "cosine", also known as a 'clock proxy' that represents the circadian rhythm (body internal clock). This was computed using the following equation 1, where t is the current time, t_0 is the time shift (5 hours [4]) and T is the period (24 hours). The values of this variable goes from -1 to +1, where negative values are night and positive values are day.

$$cosine = cos(\frac{2\pi}{T} * (t - t_0)) \tag{1}$$

6. They transform the heart rates into standard deviation of the heart rate and then scale the values.

The output data are timestamps, cosine, standard deviation of the heart rate and the sleep labels. We conducted some modifications in their preprocessing, instead of outputting only the standard deviation of the heart rate, we also modified it to output also the heart rates.

A further preprocessing to enhance the dateset was conducted by us.

We calculated the overall Motion Magnitude (from x, y, z acceleration) and its statistical properties (mean, std) over 5-minute windows. We also created cosine_xz and cosine_xy, which approximate the watch's angle on the wrist. The Motion Magnitude was obtained by making the following equation: $\sqrt{x^2 + y^2 + z^2}$. And cosine_xy is obtained by: $\cos\left(\arctan\left(\frac{y}{x}\right)\right)$ and cosine_xz by: $\cos\left(\arctan\left(\frac{z}{x}\right)\right)$. And then we interpolated their timestamps into the same timestamps as before.

We performed this feature engineering, because the activity count can miss subtle movements. So getting the distribution of the acceleration magnitude over a longer window (5 min) gives a detailed picture of movement intensity and consistency, which may help to distinguish sleep stages.

Moreover, we aggregated step data to create daily step counts, mean steps per day for each participant, and classified participants into Physical Activity Categories

(Sedentary, Low Active, etc.) based on established step thresholds by Tudor-Locke and Bassett Jr [8][9].

We performed this feature engineering to have deeper understanding about the lifestyle context as the research by Dzierzewski [1] clearly states that daily physical activity and routine are linked with overall sleep health.

After these preprocessing part, we obtain 3 different datasets: the first one obtained from the Walch [4] preprocessing the motion dataset and the steps dataset. To be able to work with them, we merged them by the user id.

The resulting dataset, each observation contains the sleep label, cardiovascular, motion, time among others signals, and also contains some lifestyle context.

Analyzing the resulting dataset, we have found out that it contains many missing values in the number of steps for the days 0, 4, 5 and 6 (which are features) before the polysomnography (PSG) starts. This is probably because some participants didn't wear the Apple Watch or there were some device errors. As the percentage of missing values were very low as shown in the Table 1, we imputed them instead of deleting the column. The imputation methods used has been KNN with 5 neighbors, because this method usually preserves patterns in the data and takes into account many other variables, as we need some contextual awareness.

Variable	Missing %
Day 0	0.070131
Day 4	0.038068
Day 5	0.107139
Day 6	0.183862

Table 1: Percentage of missing data for each of the variables that contains missing values

Finally, we performed outlier detection using a two-step method combining Principal Component Analysis (PCA) and Z-score filtering. First, we applied PCA to our feature space to reduce its dimensionality while preserving the most important patterns in the data. By projecting the data onto its principal components, we could identify many data points clearly deviated from any of the clusters, as it is shown in the figure 1.

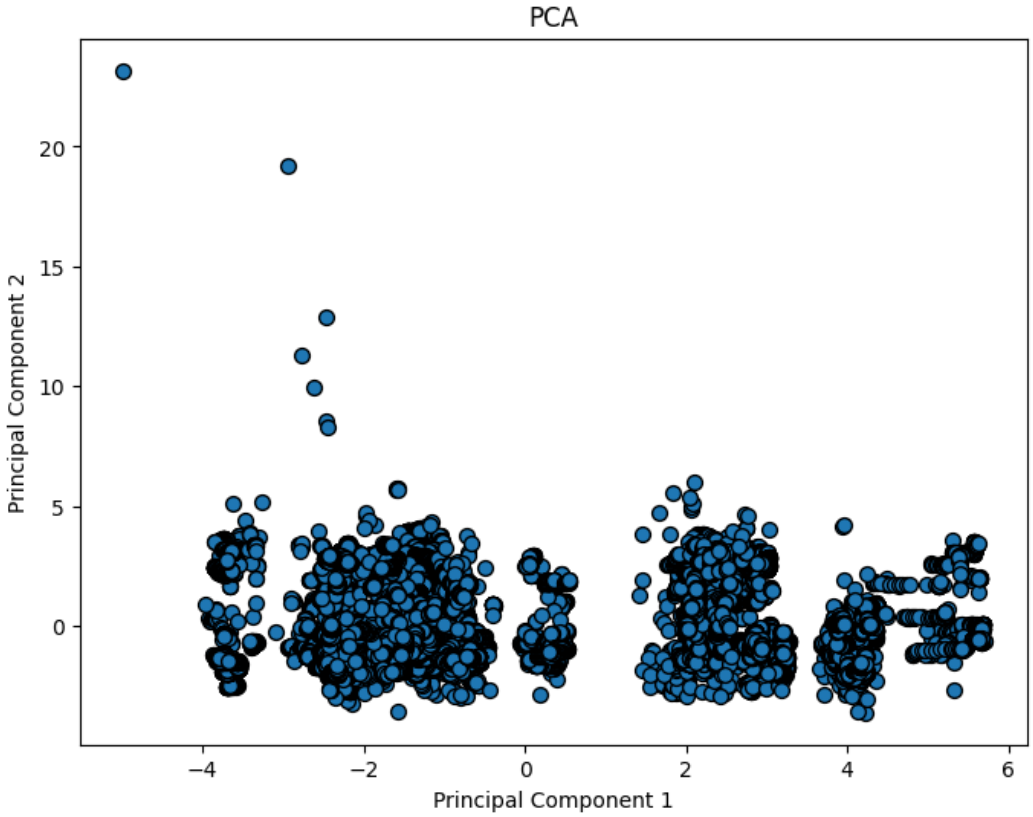


Figure 1: The projection of all samples onto Components 1 and 2 of the Principal Component Analysis (PCA)

We then computed Z-scores on the most important principal components, until the explained variance reached 80%, which are 11 components as we can see in the figure 2. Using a threshold of $|Z| > 3$, we identified which are the extreme outliers and, thus, delete them.

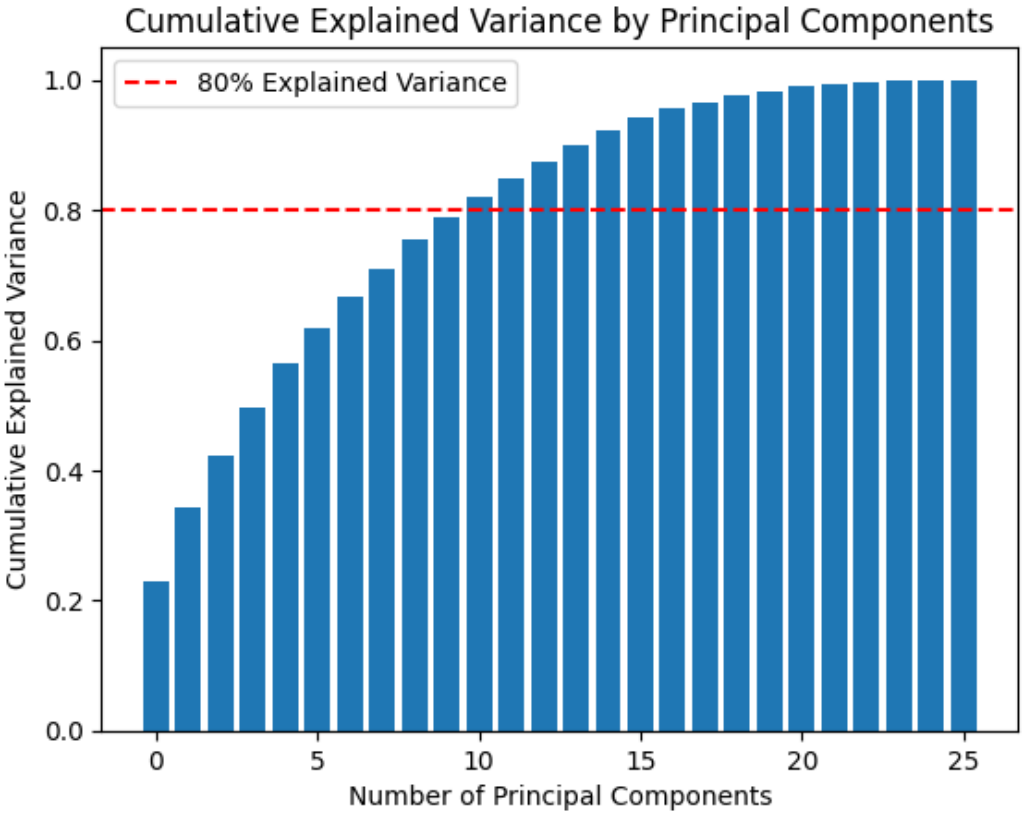


Figure 2: Cumulative Explained Variance by Principal Components

Using this two-step method, we ensured that we only removed truly anomalous records that could skew our analysis and model training. This, rather than examining each feature individually, it considered the complete relationships between all features, making it robust for complex datasets.

3.3 Data Visualization

The figure 3 provides a multi-sensor visualization for the subject 9961348 including sleep labels, acceleration, heart-rate, activity counts, and a cosine-based circadian proxy. The sleep label shows that the subject has some awakenings, many stage transitions during the early-night and longer REM periods during the late-night. Moreover, the acceleration signals remain flat during sustained sleep and show high transitions during awakenings and sleep stage changes. Additionally, activity counts corroborate the acceleration data, showing peaks aligned with with wake periods and sleep stage changes.

The heart rate panel shows that in general the heart rate decreases during any of the sleep stages except during the REM stage, which seems to have normal heart rate values, around 70 bpm. The cosine feature at the bottom represents time-of-night and goes smoothly from higher values at sleep onset to a minimum mid-sleep and back up toward morning. This feature captures circadian structure and helps models differentiate between early-night and late-night.

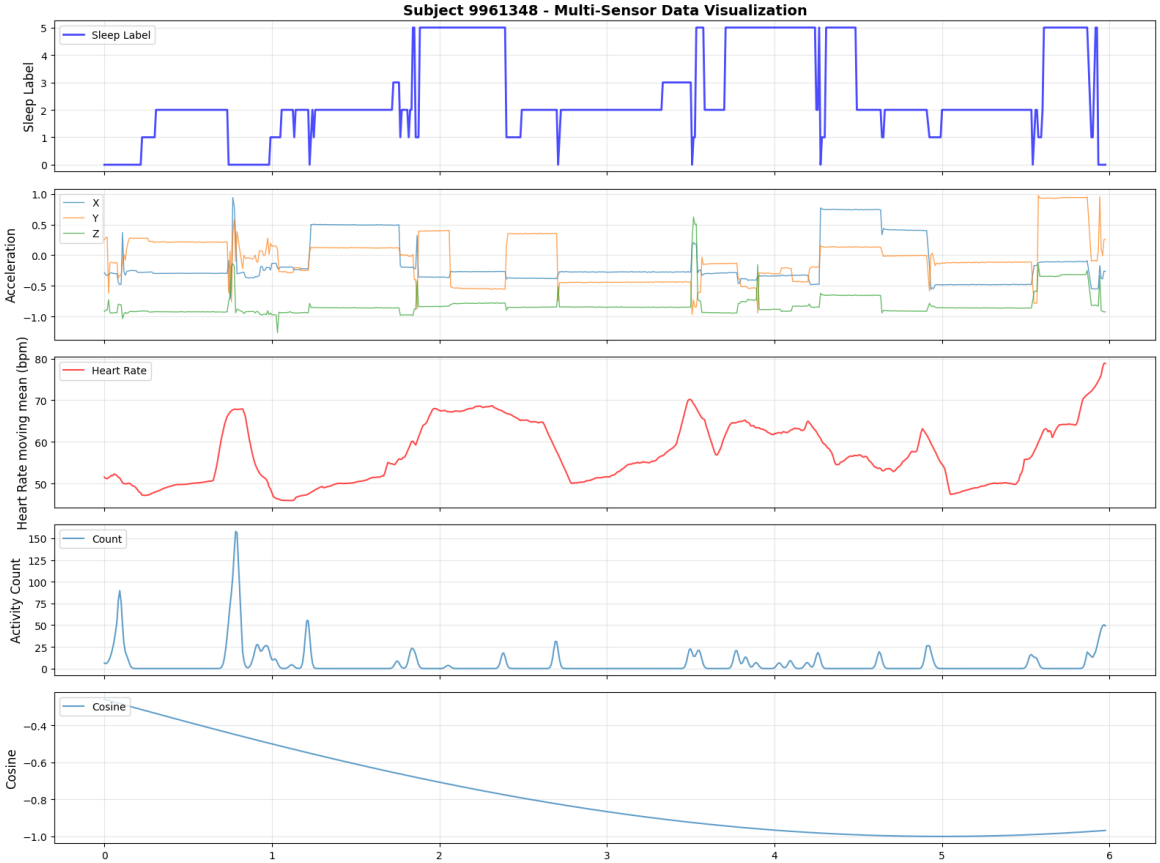


Figure 3: Multi-sensor Data Visualization for subject 9961348

The figure 4 displays the steps count over approximately seven days for the same subject as before. It reveals an irregular movement behavior. Most of the time the subject shows steps around 400, but having a extreme value exceeding 1400 steps around day 3.5. The inactive moments should match with the sleeping time.

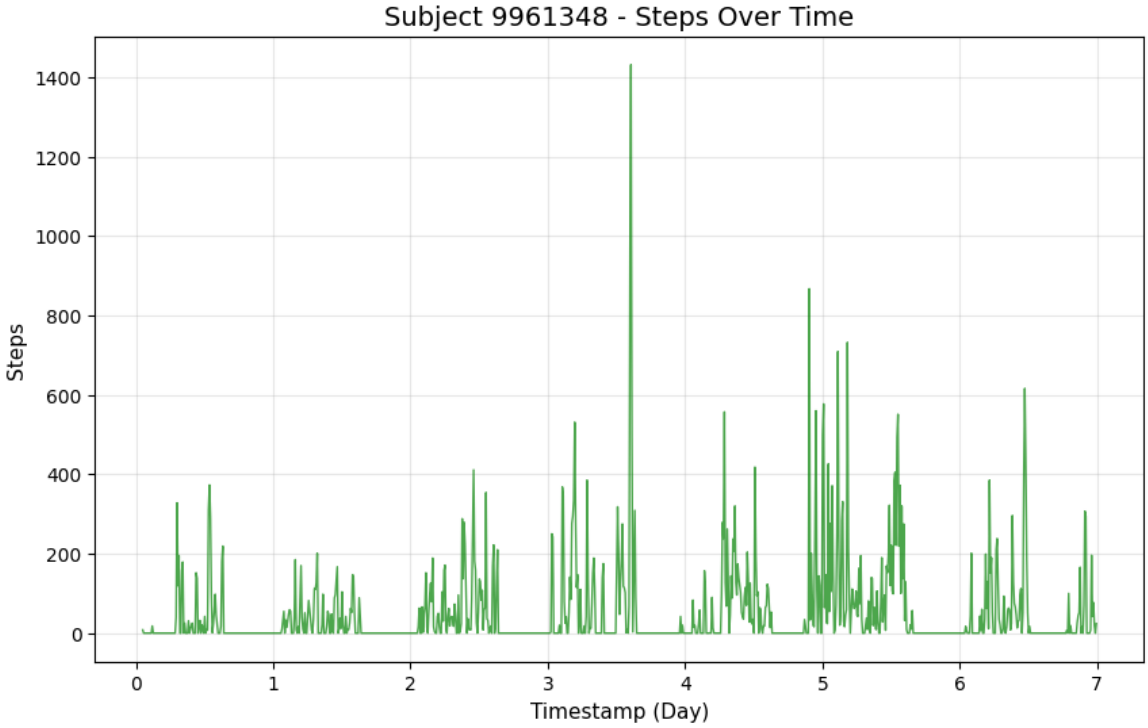


Figure 4: Steps Over time before the PSG Starts for subject 9961348

The plot 5 summarizes the relationship between the sleep label and the cosine value over the night. The stage 2 (N2) seems to be the predominant sleep label across all subjects, sometimes during the night spikes into deep sleep (N3) and REM. The stage 3 is mostly dominates the early night stages and REM increases toward morning. The cosine curve reaches its lowest point midway through the sleep period. This illustrates how circadian phase interacts with sleep. Decreasing cosine values correspond to deep sleep, while rising cosine values correspond to REM-rich later portions of the night.

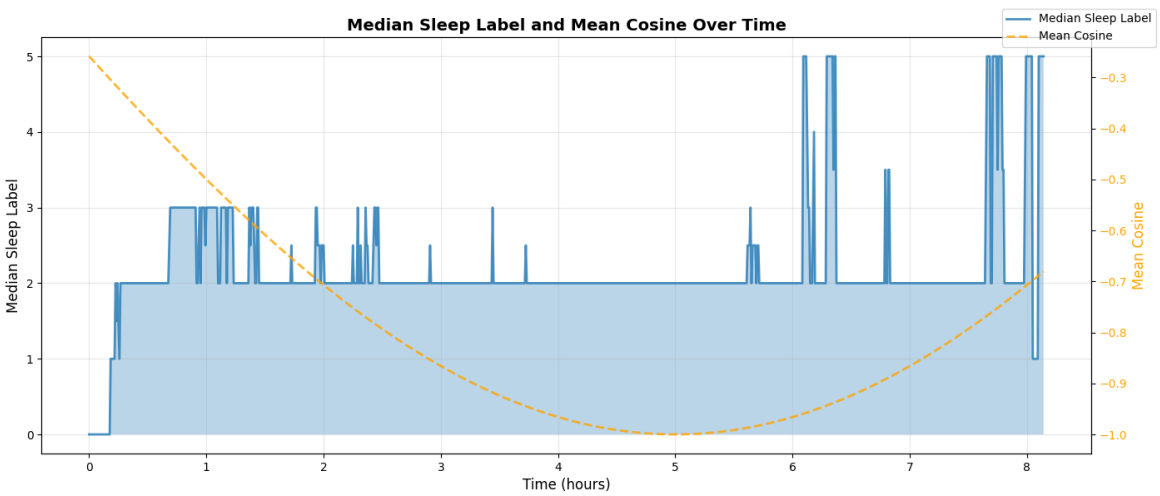


Figure 5: Median Sleep Label and Mean Cosine Over the sleeping times

4 Methods

This section explains the methods we used to analyze the data and build our models. For the data analysis we used correlation matrix and, to predict sleep stages, we tested six different model types with multiple hyperparameters settings, in order to find the best performer. We started with simpler models and moved to more complex ones.

4.1 Basic Methods

We used these simple methods to first understand our data:

- **Correlation Matrix**: We calculated Pearson correlations between all features to see how each of our features is related to every other feature. We did it to see if any of the features are highly correlated with the target and drop the column if possible, in order to avoid redundancy.
- **Logistic Regression**: We used this model as a baseline, because it is the simplest classification model. If more complex models can't beat this simple one, our features might not be good enough.

4.2 Advanced Methods

Then we went beyond these more advanced methods for classification.

- **K-Nearest Neighbors (KNN)**: We tested the KNN model with 3, 5, and 7 neighbors. This model is good for finding local patterns in sleep data.
- **Random Forest**: We tested with 100 and 200 trees. This is a powerful model that can find complex patterns and tell us which features are most important, namely provides interpretability.
- **Gradient Boosting**: We tested with 100, 200, and 500 number of estimators. This model builds trees sequentially to correct previous mistakes and often gives the best performance, and also provides interpretability.
- **Support Vector Machine (SVM)**: We tested with both rbf and poly kernels to draw boundaries between sleep stages in different ways. It could find complex boundaries that tree models might miss.

4.3 Experiment Setup

To set up the experiments, before of all, we selected a seed to be able to replicate the same results. Then we separated the dataset from the target variable. We split our data into two parts, an 80% of the data for training and a 20% of the data for testing, stratification was employed to keep proportional representation of each sleep stage in both training and testing data, in order to avoid class imbalance. It is summarized in the table 2. Then, feature scaling was then implemented, standardizing all features to have zero mean and unit variance. The evaluation consisted in four key performance metrics: accuracy (overall correctness), precision (correctness of positive predictions),

recall (ability to find all positive instances), and F1-score (harmonic mean of precision and recall). After setting up the experiments we ran all the models defined above.

Furthermore, for the best performing model, we also plotted a ROC-AUC curve and a confusion matrix, and the metrics per each class. We also created a SHAP explainer to understand the impact of the variables when predicting stage 3 (N3), in other words, the impact when predicting deep sleep.

Statistics	Training Dataset	Testing Dataset
# of rows	17104	4276
# Variables	25	25
% Stage 0	6.66 %	6.67 %
% Stage 1	6.62 %	6.62 %
% Stage 2	50.18 %	50.16 %
% Stage 3	12.77 %	12.77 %
% Stage 4	1.17 %	1.17 %
% Stage 5	22.60 %	22.61 %

Table 2: Summary statistics about the training and testing datasets

5 Results

5.1 Correlation Matrix

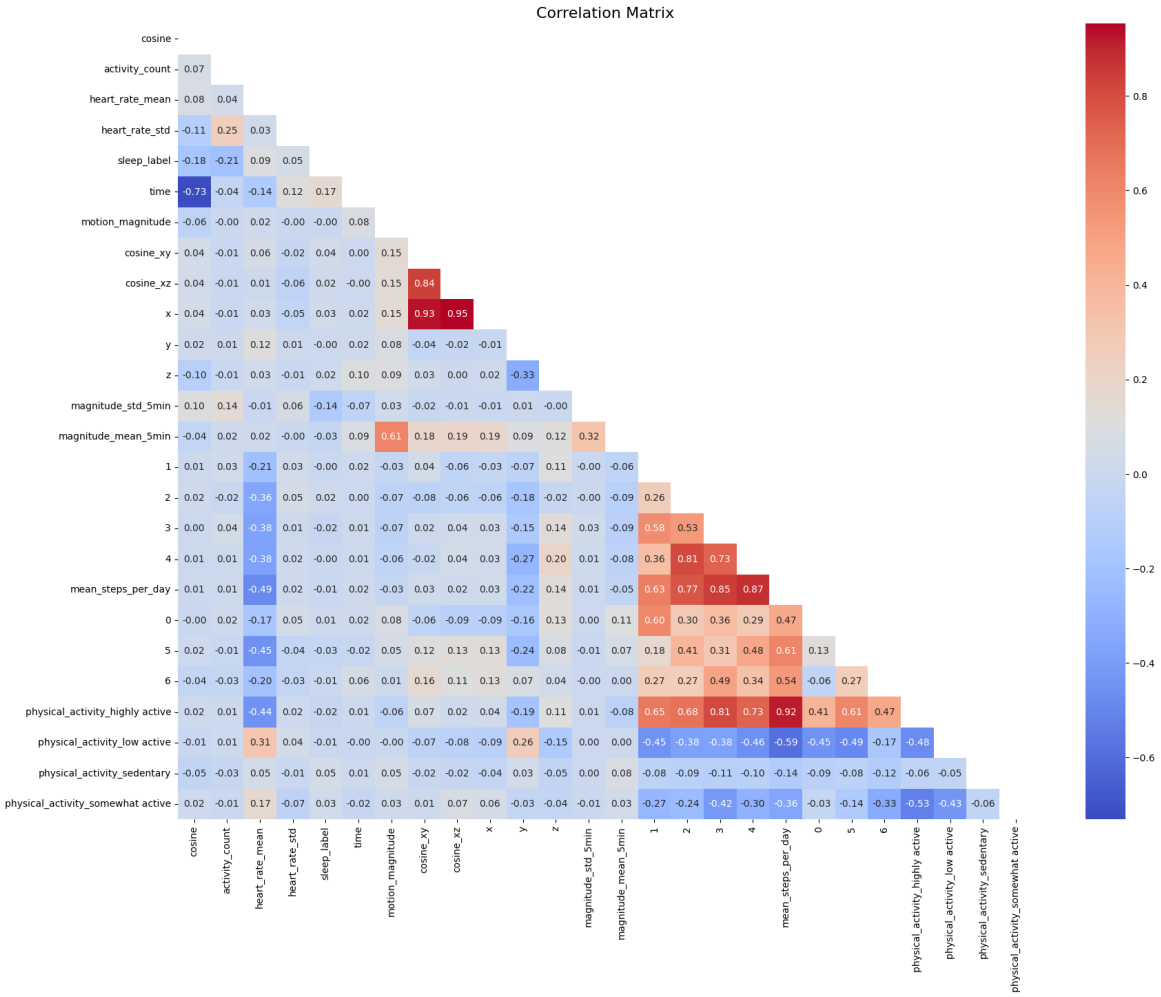


Figure 6: Correlation plot of the complete dataset

5.2 Model Comparison

After running all models, we can compare in the figure 7 how well each of the model perform on the classification task.

The Random Forest using 100 and 200 estimators are the models that stand out as the top choices, scoring very high (around 0.89 to 0.90) across all four metrics. The Gradient Boosting with 500 estimators also performs very well, but slightly lower than Random Forest either with 100 or 200 estimators.

The worst model is Logistic Regression has the lowest scores (around 0.50 to 0.55), as it was expected, because it was the baseline model.

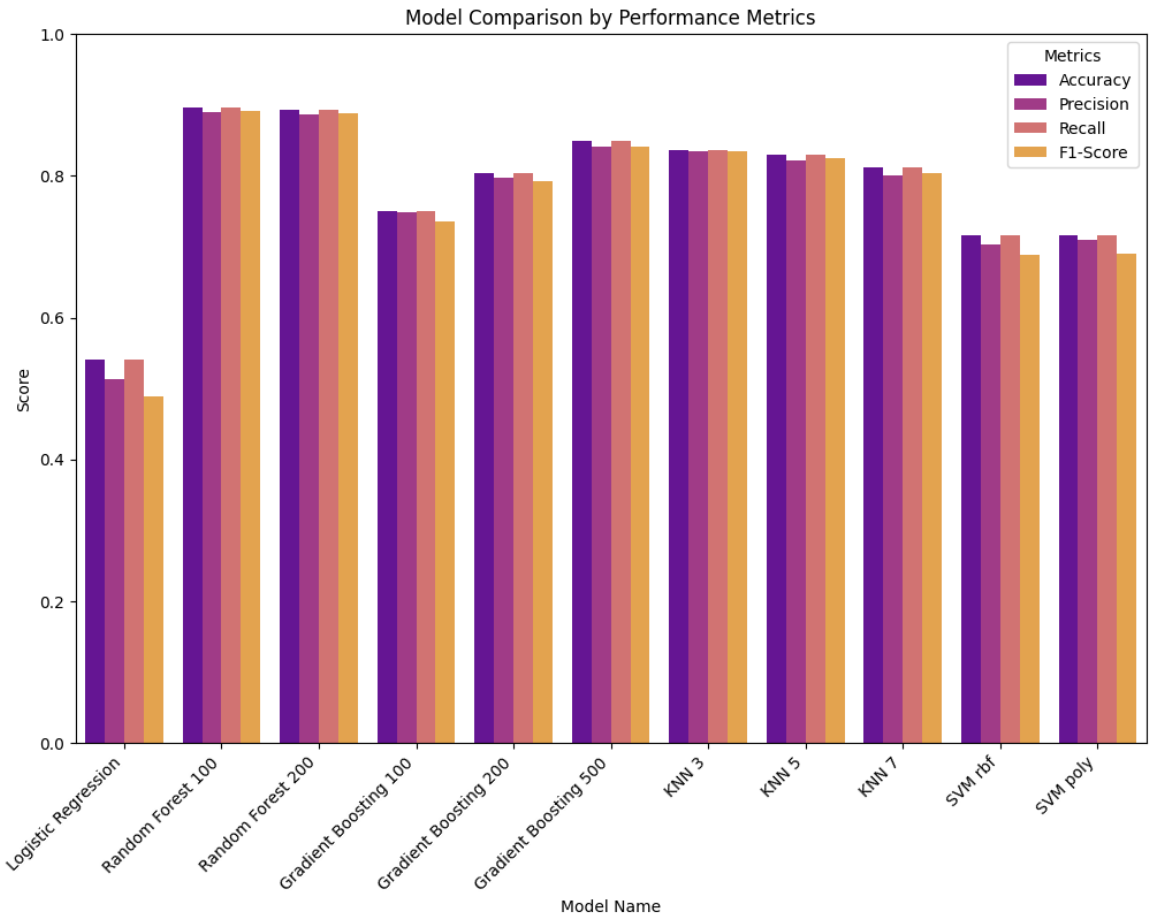


Figure 7: Model comparison by performance metrics

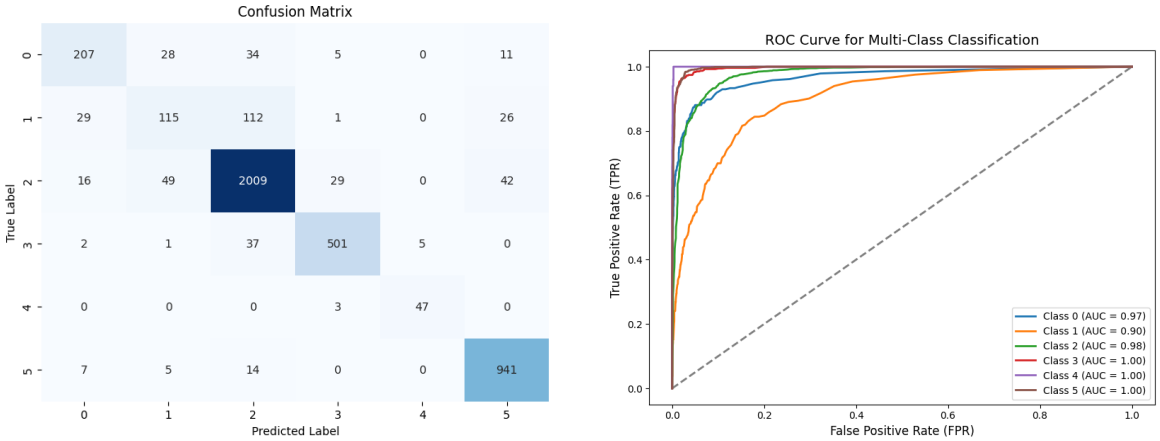
5.3 Best model Results

After comparing different models, we have chosen the best one as our final model, and this is the Random Forest with 200 estimators.

To analyze the results we plotted the confusion matrix, the ROC-AUC curve for each class and a table with the classification metrics for each class.

The overall model Accuracy is high 89% as it shows in the Table 3. The model is highly effective at classifying sleep stages from 2 to 5, which all have excellent metrics, 90% or higher in all the scores, and their AUC values in figure 8b are near-perfect (0.98 to 1.00). For example, the confusion matrix in figure 8a shows that the stage 2 (the largest class with 2145 observations) is correctly predicted 2009 times.

The model struggles on classifying the stage 1, it has the lowest scores across the sleep stages, with a precision of 58%, a recall of 41% and aF1-Score of 48%. This bad performance is also visually explained by the confusion matrix and the ROC-AUC Curve. In the confusion matrix, stage 1 is most often mistaken for stage 2 (112 incorrect predictions). On the ROC-AUC Curve, the curve for stage 1 is visibly lower than all others and has the lowest AUC (0.90).



(a) Confusion Matrix (b) ROC-AUC Curve

Figure 8: Model performance visualizations

Class	Precision	Recall	F1-score	Support
0.0	0.79	0.73	0.76	285
1.0	0.58	0.41	0.48	283
2.0	0.91	0.94	0.92	2145
3.0	0.93	0.92	0.92	546
4.0	0.90	0.94	0.92	50
5.0	0.92	0.97	0.95	967
Accuracy			0.89	4276
Macro avg	0.84	0.82	0.83	4276
Weighted avg	0.89	0.89	0.89	4276

Table 3: Classification report metrics for each class

5.3.1 Interpretability

Once we have analized the model performance and results, we perform the interpretability of the model using SHAP. The SHAP plots 9 illustrates the effect of some of the variables on the model's output for Stage 3 prediction. These plots, the y axis represent the SHAP value and the x axis represent the value in the dataset; those data points that has a SHAP value above zero means that the feature's value pushes the prediction towards Stage 3 (deep sleep), while a value below zero pushes the prediction away from Stage 3.

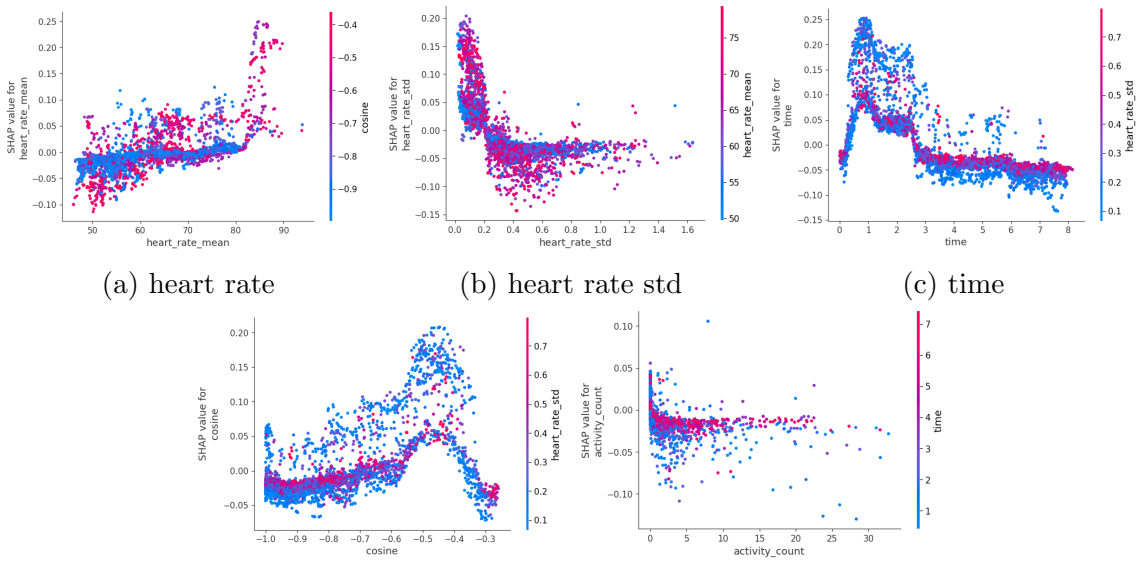
For mean heart rate in figure 9a, we cannot really see a pattern, but a higher heart rate (+80 bpm) has a big impact on predicting Stage 3, as those datapoints have positive SHAP.

For heart rate standard deviation in figure 9b, we can clearly see that more stable and less variation in the heart rate strongly contributes to predicting deep sleep, while higher variability pushes the prediction away

For time in figure 9c, the model associates very low time values (few hours after going to sleep) with high positive SHAP values, suggesting that Stage 3 deep sleep is predicted to occur early in the sleep cycle.

For activity count in figure 9e, seems that lower activity count is a key condition for predicting stage 3.

And finally, for the figure 9d, we can see that most of the positive values are around -0.6 and -0.4 cosine values. If we recall the visualization in 5, we can see that where there is more stage 3, has cosine values around -0.4 to 0.-6. The same as this SHAP values plot.



(a) heart rate (b) heart rate std (c) time

(d) cosine (e) activity count

Figure 9: Shap values dependence plots

6 Conclusion and Discussion

The primary objectives were to achieve high prediction accuracy and to identify the feature conditions that support a longer deep sleep (Stage N3) phase.

This project has successfully developed a model to accurately classify distinct sleep stages. Of the many trained models, the Random Forest model with 200 estimators achieved the highest overall accuracy (89%), demonstrating its effectiveness in classifying most stages, except for stage 1. Based on our analysis of the model predictions, we identified three characteristics for achieving deep sleep (Stage N3), which has a direct influence on overall sleep quality. The first is cardiovascular stability, low heart rate variability increases the chances of achieving deep sleep. Secondly, low activity is necessary for entering deep sleep stages. And thirdly, an optimal circadian timing is necessary, as deep sleep usually occurs during the early hours of the night (approximately one to three hours after sleeping).

In conclusion, the success of the Random Forest model demonstrates that Apple Watch data combined with some feature engineering is effective in classifying sleep stages beyond simple sleep/wake classification. SHAP values offer interpretability and provide insights into the requirements for deep sleep. And, conditions that support a longer deep sleep phase are a stable heart rate and low activity levels, with the deep sleep phase concentrated in the early hours of the night.

6.1 Limitations and Future Work

One of the primary limitations is the difficulty classifying well Stage 1 (N1). Additionally, the dataset was collected from a relatively small group of 31 healthy participants. And most critically, to ensure the identified conditions for deep sleep are universally applicable, the model should be validated by testing on a larger and diver population, as all the participants are from the University of Michigan.

For future work, in addition to improving N1/N2 discrimination, it could involve developing personalised sleep recommendations based on these physiological patterns, as well as expanding the model to take into account environmental factors such as light exposure and bedroom temperature.

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